

Assignment 1 „3d Computer Vision“

Deadline 20.04.2026

Preliminary remarks

1. Get your favorite IDE with Qt 6.10 and OpenGL running on your computer, preferably QtCreator for C++ from here <https://www.qt.io/>
 - a. Create an student account and download the Qt Open Source version for your favorite platform.
 - b. Do not forget to configure the download to your C++ compiler (MSCV, MinGW, etc.).
2. Download the **Framework** from Moodle and get it running on your computer.
 - a. The Framework contains the necessary C++ source code, two external libraries (freeglut, eigen) and some example data files.
 - b. If you use QtCreator, the file extensions `.pro` should be associated with it and QtCreator should open on double-click on the **Framework.pro** file.
3. Do **not** use functions from OpenGL, OpenCV, or any other library to compute affine and projective maps!
4. Use the template 3d-rendering **Framework** provided and edit, extend and modify appropriately.

Part 1 (3d Scene)

Use OpenGL commands to render **your own scene (consisting of wireframe models of some objects)**. This is the **green** part in **Figure 1** encircled with “scene objects in $\mathbb{R}^3$ ”. The appropriate lines in the code in `glwidget.cpp` are marked with a TODO-comment. Examples are provided in the **Framework**.

Part 2 (Perspective Camera Model)

Implement your own **perspective camera model** consisting of all intrinsic and extrinsic camera parameters and render the relevant quantities, e.g., center of projection, camera pose, image plane, image principle point, etc. This is the **red** component in **Figure 1** encircled with “perspective camera”.

Remark: The **rendering camera** class provided in the **Framework** is used for rendering purpose only. Do not change anything here. It is **NOT** the perspective camera of the assignment. This is the **blue** component in **Figure 1**.

Part 3 (Perspective Projection)

Project the scene from **Part 1** onto the image plane of the perspective camera model of **Part 2** using the perspective camera model of **Part 2**. This is the **green** component in the **image plane of the perspective camera** in **Figure 1** encircled with “projected image”.

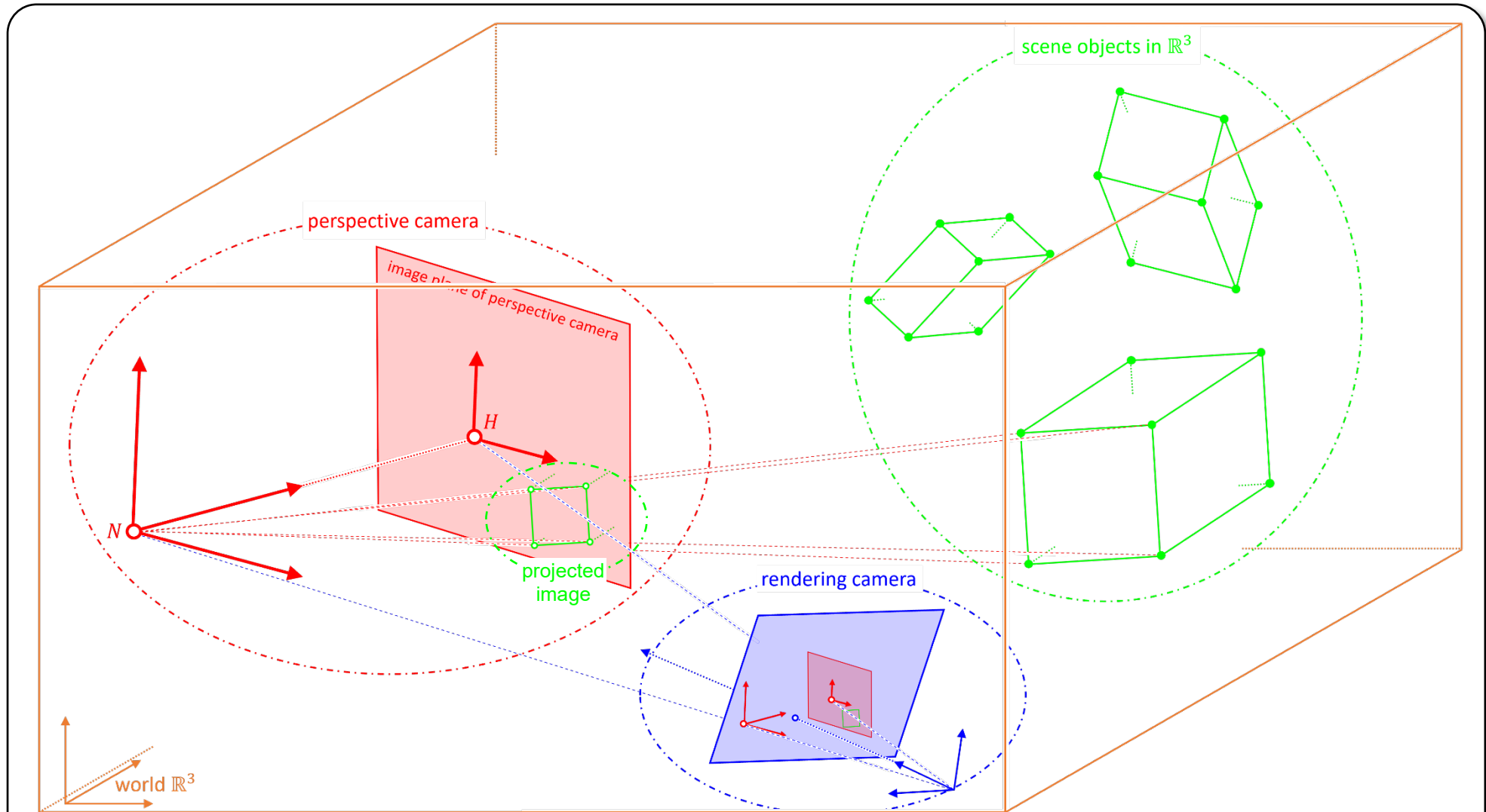


Figure 1: Schematic view of the relation between the **3d scene** (Part 1), the **perspective camera** (Part 2), the **3d scene** projected onto the **image plane of the perspective camera** (Part 3), and the **rendering camera** (included the **Framework**).